DBMS Assignment 11

Tables:

Classroom table:

```
mysql> select * from classroom;
+----------+-------------+----------+
| building | room_number | capacity |
+----------+-------------+----------+
| Packard  | 101         |      500 |
| Painter  | 514         |       10 |
| Taylor   | 3128        |       70 |
| Watson   | 100         |       30 |
| Watson   | 120         |       50 |
+----------+-------------+----------+
5 rows in set (0.01 sec)
```

Department table:

```
mysql> select * from department;
+------------+----------+-----------+
| dept_name  | building | budget    |
+------------+----------+-----------+
| Biology    | Watson   |  90000.00 |
| Comp. Sci. | Taylor   | 100000.00 |
| Elec. Eng. | Taylor   |  85000.00 |
| Finance    | Painter  | 120000.00 |
| History    | Painter  |  50000.00 |
| Music      | Packard  |  80000.00 |
| Physics    | Watson   |  70000.00 |
+------------+----------+-----------+
7 rows in set (0.00 sec)
```

Course table:

```
mysql> select * from course;
+-----------+----------------------------+------------+---------+
| course_id | title                      | dept_name  | credits |
+-----------+----------------------------+------------+---------+
| BIO-101   | Intro. to Biology          | Biology    |       4 |
| BIO-301   | Genetics                   | Biology    |       4 |
| BIO-399   | Computational Biology      | Biology    |       3 |
| CS-101    | Intro. to Computer Science | Comp. Sci. |       4 |
| CS-190    | Game Design                | Comp. Sci. |       4 |
| CS-315    | Robotics                   | Comp. Sci. |       3 |
| CS-319    | ImageProcessing            | Comp. Sci. |       3 |
| CS-347    | DatabaseSystem Concepts    | Comp. Sci. |       3 |
| EE-181    | Intro. toDigital Systems   | Elec. Eng. |       3 |
| FIN-201   | Investment Banking         | Finance    |       3 |
| HIS-351   | WorldHistory               | History    |       3 |
| MU-199    | Music VideoProduction      | Music      |       3 |
| PHY-101   | Physical Principles        | Physics    |       4 |
+-----------+----------------------------+------------+---------+
13 rows in set (0.00 sec)
```

instructor table:

```
mysql> select * from instructor;
+-------+------------+------------+----------+
| ID    | name       | dept_name  | salary   |
+-------+------------+------------+----------+
| 10101 | Srinivasan | Comp. Sci. | 65000.00 |
| 12121 | Wu         | Finance    | 90000.00 |
| 15151 | Mozart     | Music      | 40000.00 |
| 22222 | Einstein   | Physics    | 95000.00 |
| 32343 | El Said    | History    | 60000.00 |
| 33456 | Gold       | Physics    | 87000.00 |
| 58583 | Califieri  | History    | 62000.00 |
| 76543 | Singh      | Finance    | 80000.00 |
| 76766 | Crick      | Biology    | 72000.00 |
| 83821 | Brandt     | Comp. Sci. | 92000.00 |
| 98345 | Kim        | Elec. Eng. | 80000.00 |
+-------+------------+------------+----------+
11 rows in set (0.00 sec)
```

Section table:

```
mysql> select * from section;
+-----------+--------+----------+------+----------+-------------+--------------+
| course_id | sec_id | semester | year | building | room_number | time_slot_id |
+-----------+--------+----------+------+----------+-------------+--------------+
| BIO-101   | 1      | Summer   | 2009 | Painter  | 514         | B            |
| BIO-301   | 1      | Summer   | 2010 | Painter  | 514         | A            |
| CS-101    | 1      | Fall     | 2009 | Packard  | 101         | H            |
| CS-101    | 1      | Spring   | 2010 | Packard  | 101         | F            |
| CS-190    | 1      | Spring   | 2009 | Taylor   | 3128        | E            |
| CS-190    | 2      | Spring   | 2009 | Taylor   | 3128        | A            |
| CS-315    | 1      | Spring   | 2010 | Watson   | 120         | D            |
| CS-319    | 1      | Spring   | 2010 | Watson   | 100         | B            |
| CS-319    | 2      | Spring   | 2010 | Taylor   | 3128        | C            |
| CS-347    | 1      | Fall     | 2009 | Taylor   | 3128        | A            |
| EE-181    | 1      | Spring   | 2009 | Taylor   | 3128        | C            |
| FIN-201   | 1      | Spring   | 2010 | Packard  | 101         | B            |
| HIS-351   | 1      | Spring   | 2010 | Painter  | 514         | C            |
| MU-199    | 1      | Spring   | 2010 | Packard  | 101         | D            |
| PHY-101   | 1      | Fall     | 2009 | Watson   | 100         | A            |
+-----------+--------+----------+------+----------+-------------+--------------+
15 rows in set (0.00 sec)
```

Teaches table:

```
mysql> select * from teaches;
+-------+-----------+--------+----------+------+
| ID    | course_id | sec_id | semester | year |
+-------+-----------+--------+----------+------+
| 76766 | BIO-101   | 1      | Summer   | 2009 |
| 76766 | BIO-301   | 1      | Summer   | 2010 |
| 10101 | CS-101    | 1      | Fall     | 2009 |
| 83821 | CS-190    | 1      | Spring   | 2009 |
| 83821 | CS-190    | 2      | Spring   | 2009 |
| 10101 | CS-315    | 1      | Spring   | 2010 |
| 83821 | CS-319    | 2      | Spring   | 2010 |
| 10101 | CS-347    | 1      | Fall     | 2009 |
| 98345 | EE-181    | 1      | Spring   | 2009 |
| 12121 | FIN-201   | 1      | Spring   | 2010 |
| 32343 | HIS-351   | 1      | Spring   | 2010 |
| 15151 | MU-199    | 1      | Spring   | 2010 |
| 22222 | PHY-101   | 1      | Fall     | 2009 |
+-------+-----------+--------+----------+------+
13 rows in set (0.00 sec)
```

student table:

```
mysql> select * from student;
+-------+----------+------------+----------+
| ID    | name     | dept_name  | tot_cred |
+-------+----------+------------+----------+
| 00128 | Zhang    | Comp. Sci. |      102 |
| 12345 | Shankar  | Comp. Sci. |       32 |
| 19991 | Brandt   | History    |       80 |
| 23121 | Chavez   | Finance    |      110 |
| 44553 | Peltier  | Physics    |       56 |
| 45678 | Levy     | Physics    |       46 |
| 54321 | Williams | Comp. Sci. |       54 |
| 55739 | Sanchez  | Music      |       38 |
| 70557 | Snow     | Physics    |        0 |
| 76543 | Brown    | Comp. Sci. |       58 |
| 76653 | Aoi      | Elec. Eng. |       60 |
| 98765 | Bourikas | Elec. Eng. |       98 |
| 98988 | Tanaka   | Biology    |      120 |
+-------+----------+------------+----------+
13 rows in set (0.00 sec)
```

takes table:

```
mysql> select * from takes;
+-------+-----------+--------+----------+------+-------+
| ID    | course_id | sec_id | semester | year | grade |
+-------+-----------+--------+----------+------+-------+
| 00128 | CS-101    | 1      | Fall     | 2009 | A     |
| 00128 | CS-347    | 1      | Fall     | 2009 | A-    |
| 12345 | CS-101    | 1      | Fall     | 2009 | C     |
| 12345 | CS-190    | 2      | Spring   | 2009 | A     |
| 12345 | CS-315    | 1      | Spring   | 2010 | A     |
| 12345 | CS-347    | 1      | Fall     | 2009 | A     |
| 19991 | HIS-351   | 1      | Spring   | 2010 | B     |
| 23121 | FIN-201   | 1      | Spring   | 2010 | C+    |
| 44553 | PHY-101   | 1      | Fall     | 2009 | B-    |
| 45678 | CS-101    | 1      | Fall     | 2009 | F     |
| 45678 | CS-101    | 1      | Spring   | 2010 | B+    |
| 45678 | CS-319    | 1      | Spring   | 2010 | B     |
| 54321 | CS-101    | 1      | Fall     | 2009 | A-    |
| 54321 | CS-190    | 2      | Spring   | 2009 | B+    |
| 55739 | MU-199    | 1      | Spring   | 2010 | A-    |
| 76543 | CS-101    | 1      | Fall     | 2009 | A     |
| 76543 | CS-319    | 2      | Spring   | 2010 | A     |
| 76653 | EE-181    | 1      | Spring   | 2009 | C     |
| 98765 | CS-101    | 1      | Fall     | 2009 | C-    |
| 98765 | CS-315    | 1      | Spring   | 2010 | B     |
| 98988 | BIO-101   | 1      | Summer   | 2009 | A     |
| 98988 | BIO-301   | 1      | Summer   | 2010 | NULL  |
+-------+-----------+--------+----------+------+-------+
22 rows in set (0.00 sec)
```

advisor table:

```
mysql> select * from advisor;
+-------+-------+
| s_ID  | i_ID  |
+-------+-------+
| 12345 | 10101 |
| 44553 | 22222 |
| 45678 | 22222 |
| 23121 | 76543 |
| 98988 | 76766 |
| 76653 | 98345 |
| 98765 | 98345 |
+-------+-------+
7 rows in set (0.00 sec)
```

time_slot table:

```
mysql> select * from time_slot;
+--------------+-----+----------+-----------+--------+---------+
| time_slot_id | day | start_hr | start_min | end_hr | end_min |
+--------------+-----+----------+-----------+--------+---------+
| A            | F   |        8 |         0 |      8 |      50 |
| A            | M   |        8 |         0 |      8 |      50 |
| A            | W   |        8 |         0 |      8 |      50 |
| B            | F   |        9 |         0 |      9 |      50 |
| B            | M   |        9 |         0 |      9 |      50 |
| B            | W   |        9 |         0 |      9 |      50 |
| C            | F   |       11 |         0 |     11 |      50 |
| C            | M   |       11 |         0 |     11 |      50 |
| C            | W   |       11 |         0 |     11 |      50 |
| D            | F   |       13 |         0 |     13 |      50 |
| D            | M   |       13 |         0 |     13 |      50 |
| D            | W   |       13 |         0 |     13 |      50 |
| E            | R   |       10 |        30 |     11 |      45 |
| E            | T   |       10 |        30 |     11 |      45 |
| F            | R   |       14 |        30 |     15 |      45 |
| F            | T   |       14 |        30 |     15 |      45 |
| G            | F   |       16 |         0 |     16 |      50 |
| G            | M   |       16 |         0 |     16 |      50 |
| G            | W   |       16 |         0 |     16 |      50 |
| H            | W   |       10 |         0 |     12 |      30 |
+--------------+-----+----------+-----------+--------+---------+
20 rows in set (0.00 sec)
```

prereq table:

```
mysql> select * from prereq;
+-----------+-----------+
| course_id | prereq_id |
+-----------+-----------+
| BIO-301   | BIO-101   |
| BIO-399   | BIO-101   |
| CS-190    | CS-101    |
| CS-315    | CS-101    |
| CS-319    | CS-101    |
| CS-347    | CS-101    |
| EE-181    | PHY-101   |
+-----------+-----------+
7 rows in set (0.00 sec)
```

Question1:

a)

```
mysql> SELECT title
    -> FROM course
    -> WHERE dept_name = 'Comp. Sci.'
    ->   AND credits = 3;
+--------------------------+
| title                    |
+--------------------------+
| Robotics                 |
| Image
Processing         |
| Database
System Concepts |
+--------------------------+
3 rows in set (0.01 sec)
```

b)

```
mysql> SELECT DISTINCT t.ID
    -> FROM takes t
    -> JOIN teaches te ON t.course_id = te.course_id
    ->                AND t.sec_id = te.sec_id
    ->                AND t.semester = te.semester
    ->                AND t.year = te.year
    -> JOIN instructor i ON te.ID = i.ID
    -> WHERE i.name = 'Einstein';
+-------+
| ID    |
+-------+
| 44553 |
+-------+
```

c)

```
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM instructor;
+-----------------+
| highest_salary |
+-----------------+
|       95000.00 |
+-----------------+
1 row in set (0.01 sec)
```

d)

```
mysql> SELECT ID, name, salary
    -> FROM instructor
    -> WHERE salary = (SELECT MAX(salary) FROM instructor);
+-------+----------+----------+
| ID    | name     | salary   |
+-------+----------+----------+
| 22222 | Einstein | 95000.00 |
+-------+----------+----------+
1 row in set (0.00 sec)
```

e)

```
mysql> SELECT s.course_id, s.sec_id, COUNT(t.ID) AS enrollment
    -> FROM section s
    -> JOIN takes t ON s.course_id = t.course_id
    ->              AND s.sec_id = t.sec_id
    ->              AND s.semester = t.semester
    ->              AND s.year = t.year
    -> WHERE s.semester = 'Fall'
    ->   AND s.year = 2009
    -> GROUP BY s.course_id, s.sec_id;
+-----------+--------+------------+
| course_id | sec_id | enrollment |
+-----------+--------+------------+
| CS-101    | 1      |          6 |
| CS-347    | 1      |          2 |
| PHY-101   | 1      |          1 |
+-----------+--------+------------+
3 rows in set (0.01 sec)
```

f)

```
mysql> SELECT MAX(enrollment) AS max_enrollment
    -> FROM (
    ->     SELECT COUNT(t.ID) AS enrollment
    ->     FROM section s
    ->     JOIN takes t ON s.course_id = t.course_id
    ->                 AND s.sec_id = t.sec_id
    ->                 AND s.semester = t.semester
    ->                 AND s.year = t.year
    ->     WHERE s.semester = 'Fall'
    ->       AND s.year = 2009
    ->     GROUP BY s.course_id, s.sec_id
    -> ) AS section_enrollments;
+----------------+
| max_enrollment |
+----------------+
|              6 |
+----------------+
1 row in set (0.01 sec)
```

g)

```
mysql> SELECT s.course_id, s.sec_id, COUNT(t.ID) AS enrollment
    -> FROM section s
    -> JOIN takes t ON s.course_id = t.course_id
    ->              AND s.sec_id = t.sec_id
    ->              AND s.semester = t.semester
    ->              AND s.year = t.year
    -> WHERE s.semester = 'Fall'
    ->   AND s.year = 2009
    -> GROUP BY s.course_id, s.sec_id
    -> HAVING COUNT(t.ID) = (
    ->     SELECT MAX(enrollment) AS max_enrollment
    ->     FROM (
    ->         SELECT COUNT(t.ID) AS enrollment
    ->         FROM section s
    ->         JOIN takes t ON s.course_id = t.course_id
    ->                     AND s.sec_id = t.sec_id
    ->                     AND s.semester = t.semester
    ->                     AND s.year = t.year
    ->         WHERE s.semester = 'Fall'
    ->           AND s.year = 2009
    ->         GROUP BY s.course_id, s.sec_id
    ->     ) AS section_enrollments
    -> );
+-----------+--------+------------+
| course_id | sec_id | enrollment |
+-----------+--------+------------+
| CS-101    | 1      |          6 |
+-----------+--------+------------+
1 row in set (0.00 sec)
```

Question2:

a)

```
mysql> UPDATE instructor
    -> SET salary = salary * 1.10
    -> WHERE dept_name = 'Comp. Sci.';
Query OK, 2 rows affected (0.02 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

b)

```
mysql> DELETE FROM course
    -> WHERE course_id NOT IN (SELECT DISTINCT course_id FROM section);
Query OK, 1 row affected (0.01 sec)
```

c)

```
mysql> INSERT INTO instructor (ID, name, dept_name, salary)
    -> SELECT ID, name, dept_name, 30000
    -> FROM student
    -> WHERE tot_cred > 100;
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Question3:

```
mysql> SELECT dept_name
    -> FROM department
    -> WHERE LOWER(dept_name) LIKE '%sci%';
+------------+
| dept_name  |
+------------+
| Comp. Sci. |
+------------+
1 row in set (0.01 sec)
```

Question 4:

a)

```
mysql> SELECT i.ID, i.name, COUNT(t.course_id) AS num_sections
    -> FROM instructor i
    -> LEFT OUTER JOIN teaches t ON i.ID = t.ID
    -> GROUP BY i.ID, i.name;
+-------+------------+--------------+
| ID    | name       | num_sections |
+-------+------------+--------------+
| 00128 | Zhang      |            0 |
| 10101 | Srinivasan |            3 |
| 12121 | Wu         |            1 |
| 15151 | Mozart     |            1 |
| 22222 | Einstein   |            1 |
| 23121 | Chavez     |            0 |
| 32343 | El Said    |            1 |
| 33456 | Gold       |            0 |
| 58583 | Califieri  |            0 |
| 76543 | Singh      |            0 |
| 76766 | Crick      |            2 |
| 83821 | Brandt     |            3 |
| 98345 | Kim        |            1 |
| 98988 | Tanaka     |            0 |
+-------+------------+--------------+
14 rows in set (0.00 sec)
```

b)

```
mysql> SELECT i.ID, i.name,
    ->        (SELECT COUNT(*)
    ->         FROM teaches t
    ->         WHERE t.ID = i.ID) AS num_sections
    -> FROM instructor i;
+-------+------------+--------------+
| ID    | name       | num_sections |
+-------+------------+--------------+
| 00128 | Zhang      |            0 |
| 10101 | Srinivasan |            3 |
| 12121 | Wu         |            1 |
| 15151 | Mozart     |            1 |
| 22222 | Einstein   |            1 |
| 23121 | Chavez     |            0 |
| 32343 | El Said    |            1 |
| 33456 | Gold       |            0 |
| 58583 | Califieri  |            0 |
| 76543 | Singh      |            0 |
| 76766 | Crick      |            2 |
| 83821 | Brandt     |            3 |
| 98345 | Kim        |            1 |
| 98988 | Tanaka     |            0 |
+-------+------------+--------------+
14 rows in set (0.00 sec)
```

c)

```
mysql> SELECT s.course_id, s.sec_id, s.semester, s.year,
    ->        COALESCE(i.name, '-') AS instructor_name
    -> FROM section s
    -> LEFT JOIN teaches t ON s.course_id = t.course_id
    ->                    AND s.sec_id = t.sec_id
    ->                    AND s.semester = t.semester
    ->                    AND s.year = t.year
    -> LEFT JOIN instructor i ON t.ID = i.ID
    -> WHERE s.semester = 'Spring' AND s.year = 2010;
+-----------+--------+----------+------+-----------------+
| course_id | sec_id | semester | year | instructor_name |
+-----------+--------+----------+------+-----------------+
| CS-101    | 1      | Spring   | 2010 | ?               |
| FIN-201   | 1      | Spring   | 2010 | Wu              |
| MU-199    | 1      | Spring   | 2010 | Mozart          |
| HIS-351   | 1      | Spring   | 2010 | El Said         |
| CS-319    | 2      | Spring   | 2010 | Brandt          |
| CS-319    | 1      | Spring   | 2010 | ?               |
| CS-315    | 1      | Spring   | 2010 | Srinivasan      |
+-----------+--------+----------+------+-----------------+
7 rows in set (0.00 sec)
```

d)

```
mysql> SELECT d.dept_name, COUNT(i.ID) AS num_instructors
    -> FROM department d
    -> LEFT JOIN instructor i ON d.dept_name = i.dept_name
    -> GROUP BY d.dept_name;
+------------+-----------------+
| dept_name  | num_instructors |
+------------+-----------------+
| Biology    |               2 |
| Comp. Sci. |               3 |
| Elec. Eng. |               1 |
| Finance    |               3 |
| History    |               2 |
| Music      |               1 |
| Physics    |               2 |
+------------+-----------------+
7 rows in set (0.00 sec)
```

Question 5:

a)

```
mysql> SELECT s.ID, s.name, s.dept_name, s.tot_cred, t.course_id, t.sec_id, t.semester, t.year, t.grade
    -> FROM student s
    -> LEFT JOIN takes t ON s.ID = t.ID;
+-------+----------+------------+----------+-----------+--------+----------+------+-------+
| ID    | name     | dept_name  | tot_cred | course_id | sec_id | semester | year | grade |
+-------+----------+------------+----------+-----------+--------+----------+------+-------+
| 00128 | Zhang    | Comp. Sci. |      102 | CS-101    | 1      | Fall     | 2009 | A     |
| 00128 | Zhang    | Comp. Sci. |      102 | CS-347    | 1      | Fall     | 2009 | A-    |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-101    | 1      | Fall     | 2009 | C     |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-190    | 2      | Spring   | 2009 | A     |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-315    | 1      | Spring   | 2010 | A     |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-347    | 1      | Fall     | 2009 | A     |
| 19991 | Brandt   | History    |       80 | HIS-351   | 1      | Spring   | 2010 | B     |
| 23121 | Chavez   | Finance    |      110 | FIN-201   | 1      | Spring   | 2010 | C+    |
| 44553 | Peltier  | Physics    |       56 | PHY-101   | 1      | Fall     | 2009 | B-    |
| 45678 | Levy     | Physics    |       46 | CS-101    | 1      | Fall     | 2009 | F     |
| 45678 | Levy     | Physics    |       46 | CS-101    | 1      | Spring   | 2010 | B+    |
| 45678 | Levy     | Physics    |       46 | CS-319    | 1      | Spring   | 2010 | B     |
| 54321 | Williams | Comp. Sci. |       54 | CS-101    | 1      | Fall     | 2009 | A-    |
| 54321 | Williams | Comp. Sci. |       54 | CS-190    | 2      | Spring   | 2009 | B+    |
| 55739 | Sanchez  | Music      |       38 | MU-199    | 1      | Spring   | 2010 | A-    |
| 70557 | Snow     | Physics    |        0 | NULL      | NULL   | NULL     | NULL | NULL  |
| 76543 | Brown    | Comp. Sci. |       58 | CS-101    | 1      | Fall     | 2009 | A     |
| 76543 | Brown    | Comp. Sci. |       58 | CS-319    | 2      | Spring   | 2010 | A     |
| 76653 | Aoi      | Elec. Eng. |       60 | EE-181    | 1      | Spring   | 2009 | C     |
| 98765 | Bourikas | Elec. Eng. |       98 | CS-101    | 1      | Fall     | 2009 | C-    |
| 98765 | Bourikas | Elec. Eng. |       98 | CS-315    | 1      | Spring   | 2010 | B     |
| 98988 | Tanaka   | Biology    |      120 | BIO-101   | 1      | Summer   | 2009 | A     |
| 98988 | Tanaka   | Biology    |      120 | BIO-301   | 1      | Summer   | 2010 | NULL  |
+-------+----------+------------+----------+-----------+--------+----------+------+-------+
23 rows in set (0.00 sec)
```

b)

```
mysql> SELECT s.ID, s.name, s.dept_name, s.tot_cred, t.course_id, t.sec_id, t.semester, t.year, t.grade
    -> FROM student s
    -> LEFT JOIN takes t ON s.ID = t.ID
    -> UNION
    -> SELECT s.ID, s.name, s.dept_name, s.tot_cred, t.course_id, t.sec_id, t.semester, t.year, t.grade
    -> FROM takes t
    -> LEFT JOIN student s ON t.ID = s.ID;
+-------+----------+------------+----------+-----------+--------+----------+------+-------+
| ID    | name     | dept_name  | tot_cred | course_id | sec_id | semester | year | grade |
+-------+----------+------------+----------+-----------+--------+----------+------+-------+
| 00128 | Zhang    | Comp. Sci. |      102 | CS-101    | 1      | Fall     | 2009 | A     |
| 00128 | Zhang    | Comp. Sci. |      102 | CS-347    | 1      | Fall     | 2009 | A-    |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-101    | 1      | Fall     | 2009 | C     |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-190    | 2      | Spring   | 2009 | A     |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-315    | 1      | Spring   | 2010 | A     |
| 12345 | Shankar  | Comp. Sci. |       32 | CS-347    | 1      | Fall     | 2009 | A     |
| 19991 | Brandt   | History    |       80 | HIS-351   | 1      | Spring   | 2010 | B     |
| 23121 | Chavez   | Finance    |      110 | FIN-201   | 1      | Spring   | 2010 | C+    |
| 44553 | Peltier  | Physics    |       56 | PHY-101   | 1      | Fall     | 2009 | B-    |
| 45678 | Levy     | Physics    |       46 | CS-101    | 1      | Fall     | 2009 | F     |
| 45678 | Levy     | Physics    |       46 | CS-101    | 1      | Spring   | 2010 | B+    |
| 45678 | Levy     | Physics    |       46 | CS-319    | 1      | Spring   | 2010 | B     |
| 54321 | Williams | Comp. Sci. |       54 | CS-101    | 1      | Fall     | 2009 | A-    |
| 54321 | Williams | Comp. Sci. |       54 | CS-190    | 2      | Spring   | 2009 | B+    |
| 55739 | Sanchez  | Music      |       38 | MU-199    | 1      | Spring   | 2010 | A-    |
| 70557 | Snow     | Physics    |        0 | NULL      | NULL   | NULL     | NULL | NULL  |
| 76543 | Brown    | Comp. Sci. |       58 | CS-101    | 1      | Fall     | 2009 | A     |
| 76543 | Brown    | Comp. Sci. |       58 | CS-319    | 2      | Spring   | 2010 | A     |
| 76653 | Aoi      | Elec. Eng. |       60 | EE-181    | 1      | Spring   | 2009 | C     |
| 98765 | Bourikas | Elec. Eng. |       98 | CS-101    | 1      | Fall     | 2009 | C-    |
| 98765 | Bourikas | Elec. Eng. |       98 | CS-315    | 1      | Spring   | 2010 | B     |
| 98988 | Tanaka   | Biology    |      120 | BIO-101   | 1      | Summer   | 2009 | A     |
| 98988 | Tanaka   | Biology    |      120 | BIO-301   | 1      | Summer   | 2010 | NULL  |
+-------+----------+------------+----------+-----------+--------+----------+------+-------+
23 rows in set (0.01 sec)
```

Question 6:

```
mysql> CREATE TABLE BankCustomers (
    ->     accNum INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     loan DECIMAL(15, 2) CHECK (loan <= 1000000)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER loan_check_before_insert
    -> BEFORE INSERT ON BankCustomers
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.loan > 1000000 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Loan amount cannot exceed 10 lakhs.';
    ->     END IF;
    -> END $$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER loan_check_before_update
    -> BEFORE UPDATE ON BankCustomers
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.loan > 1000000 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Loan amount cannot exceed 10 lakhs.';
    ->     END IF;
    -> END $$
Query OK, 0 rows affected (0.01 sec)
```

Verification:

```
mysql>
mysql> DELIMITER ;
mysql> INSERT INTO BankCustomers (accNum, name, loan)
    -> VALUES (1, 'John Doe', 500000);  -- Loan is within the valid range
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO BankCustomers (accNum, name, loan)
    -> VALUES (2, 'Jane Smith', 1500000);  -- Loan exceeds 10 lakhs
ERROR 1644 (45000): Loan amount cannot exceed 10 lakhs.
mysql>
```

Question 7:

```
mysql> CREATE TABLE BankingCustomers (
    ->     accNum INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     loan DECIMAL(12, 2),
    ->     major BOOLEAN  -- Assuming major is a boolean indicating if the customer is a major
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE InsertCustomerAndListMajors(
    ->     IN p_accNum INT,
    ->     IN p_name VARCHAR(100),
    ->     IN p_loan DECIMAL(12, 2),
    ->     IN p_major BOOLEAN
    -> )
    -> BEGIN
    ->     -- Declare a handler for duplicate key errors
    ->     DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
    ->     BEGIN
    ->         -- If duplicate key occurs, exit the procedure
    ->         SELECT 'Duplicate Key Error: Customer already exists. Terminating procedure.';
    ->         RESIGNAL;  -- Rethrow the exception to stop execution of the procedure
    ->     END;
    ->
    ->     -- Try to insert a new customer into the BankingCustomers table
    ->     INSERT INTO BankingCustomers (accNum, name, loan, major)
    ->     VALUES (p_accNum, p_name, p_loan, p_major);
    ->
    ->     -- Count and display the number of customers who are majors
    ->     SELECT COUNT(*) AS NumberOfMajors
    ->     FROM BankingCustomers
    ->     WHERE major = TRUE;
    ->
    -> END $$
Query OK, 0 rows affected (0.03 sec)
```

Calling stored procedure:

```
mysql>
mysql> DELIMITER ;
mysql> CALL InsertCustomerAndListMajors(101, 'Alice Green', 500000.00, TRUE);
+----------------+
| NumberOfMajors |
+----------------+
|              1 |
+----------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> CALL InsertCustomerAndListMajors(101, 'John Doe', 400000.00, FALSE);
+-----------------------------------------------------------------------+
| Duplicate Key Error: Customer already exists. Terminating procedure. |
+-----------------------------------------------------------------------+
| Duplicate Key Error: Customer already exists. Terminating procedure. |
+-----------------------------------------------------------------------+
1 row in set (0.00 sec)

ERROR 1062 (23000): Duplicate entry '101' for key 'bankingcustomers.PRIMARY'
mysql> CALL InsertCustomerAndListMajors(102, 'Bob Black', 700000.00, TRUE);
+----------------+
| NumberOfMajors |
+----------------+
|              2 |
+----------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

Question 8:

Tables:

```
mysql> CREATE TABLE Branch (
    ->     branch_id INT PRIMARY KEY,
    ->     branch_name VARCHAR(100),
    ->     total_asset DECIMAL(12, 2)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Depositor (
    ->     accNum INT,
    ->     customer_id INT,
    ->     PRIMARY KEY (accNum, customer_id),
    ->     FOREIGN KEY (accNum) REFERENCES BankingCustomers(accNum)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DELIMITER ;
mysql> CREATE TABLE BankingCustomersNew (
    ->     accNum INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     loan DECIMAL(15, 2),
    ->     branch_id INT,  -- Assuming each customer belongs to a branch
    ->     balance DECIMAL(15, 2),
    ->     address VARCHAR(255),
    ->     FOREIGN KEY (branch_id) REFERENCES Branch(branch_id)  -- Foreign Key linking to Branch table
    -> );
Query OK, 0 rows affected (0.04 sec)
```

Values in tables:

```
mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Branch (branch_id, branch_name, total_asset)
    -> VALUES
    -> (1, 'Main Branch', 5000000.00),
    -> (2, 'North Branch', 3000000.00);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO BankingCustomersNew (accNum, name, loan, branch_id, balance, address)
    -> VALUES
    -> (101, 'Alice Johnson', 100000.00, 1, 5000.00, '123 Main St'),
    -> (102, 'Bob Smith', 200000.00, 1, 15000.00, '456 Oak Ave'),
    -> (103, 'Charlie Brown', 150000.00, 2, 20000.00, '789 Pine Rd');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Trigger:

```
mysql> DELIMITER ;
mysql> CREATE TABLE BankingCustomersNew (
    ->     accNum INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     loan DECIMAL(15, 2),
    ->     branch_id INT,  -- Assuming each customer belongs to a branch
    ->     balance DECIMAL(15, 2),
    ->     address VARCHAR(255),
    ->     FOREIGN KEY (branch_id) REFERENCES Branch(branch_id)  -- Foreign Key linking to Branch table
    -> );
Query OK, 0 rows affected (0.04 sec)
```

Verification:

```
mysql> DELETE FROM BankingCustomersNew WHERE accNum = 101;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Branch;
+-----------+--------------+-------------+
| branch_id | branch_name  | total_asset |
+-----------+--------------+-------------+
|         1 | Main Branch  |  4900000.00 |
|         2 | North Branch |  3000000.00 |
+-----------+--------------+-------------+
2 rows in set (0.00 sec)
```

Question 9:

Tables:

```
mysql> CREATE TABLE Employee (
    ->     Emp_id INT PRIMARY KEY,
    ->     Emp_name VARCHAR(100),
    ->     experience INT  -- years of experience
    -> );
ERROR 1050 (42S01): Table 'employee' already exists
mysql> CREATE TABLE employeetable (
    ->     Emp_id INT PRIMARY KEY,
    ->     Emp_name VARCHAR(100),
    ->     experience INT  -- years of experience
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE EmployeePosition (
    ->     Emp_id INT PRIMARY KEY,
    ->     Emp_name VARCHAR(100),
    ->     Position VARCHAR(50),
    ->     FOREIGN KEY (Emp_id) REFERENCES employeetable(Emp_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO employeetable (Emp_id, Emp_name, experience)
    -> VALUES
    -> (1, 'Alice', 30),
    -> (2, 'Bob', 15),
    -> (3, 'Charlie', 8),
    -> (4, 'David', 3);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

Stored function:

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE AssignPositionsToAllEmployees()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE emp_id INT;
    ->     DECLARE emp_name VARCHAR(100);
    ->     DECLARE emp_experience INT;
    ->     DECLARE cur CURSOR FOR
    ->         SELECT Emp_id, Emp_name, experience FROM employeetable;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     read_loop: LOOP
    ->         FETCH cur INTO emp_id, emp_name, emp_experience;
    ->
    ->         IF done THEN
    ->             LEAVE read_loop;
    ->         END IF;
    ->
    ->         -- Debug check to see the fetched values
    ->         IF emp_id IS NULL OR emp_name IS NULL OR emp_experience IS NULL THEN
    ->             SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Null value encountered during cursor fetch';
    ->         END IF;
    ->
    ->         -- Insert into EmployeePosition table using the AssignPosition function
    ->         INSERT INTO EmployeePosition (Emp_id, Emp_name, Position)
    ->         VALUES (emp_id, emp_name, AssignPosition(emp_experience));
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    -> END $$
Query OK, 0 rows affected (0.01 sec)
```

Question 10:

Trigger:

```
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER UpdateEmployeePositionAfterInsert
    -> AFTER INSERT ON EmployeeTable
    -> FOR EACH ROW
    -> BEGIN
    ->     -- Determine the position based on the experience of the new employee
    ->     DECLARE position VARCHAR(50);
    ->
    ->     IF NEW.experience > 25 THEN
    ->         SET position = 'Senior Executive';
    ->     ELSEIF NEW.experience > 10 THEN
    ->         SET position = 'Executive';
    ->     ELSEIF NEW.experience > 5 THEN
    ->         SET position = 'Senior Analyst';
    ->     ELSE
    ->         SET position = 'Analyst';
    ->     END IF;
    ->
    ->     -- Insert the new employee's position into the EmployeePosition table
    ->     INSERT INTO EmployeePosition (Emp_id, Emp_name, Position)
    ->     VALUES (NEW.Emp_id, NEW.Emp_name, position);
    -> END $$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

Verification:

```
mysql> -- Insert a new employee into the EmployeeTable
mysql> INSERT INTO EmployeeTable (Emp_id, Emp_name, experience)
    -> VALUES (101, 'John Doe', 12);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> -- Check the EmployeePosition table
mysql> SELECT * FROM EmployeePosition;
+--------+----------+-----------+
| Emp_id | Emp_name | Position  |
+--------+----------+-----------+
|    101 | John Doe | Executive |
+--------+----------+-----------+
1 row in set (0.00 sec)
```